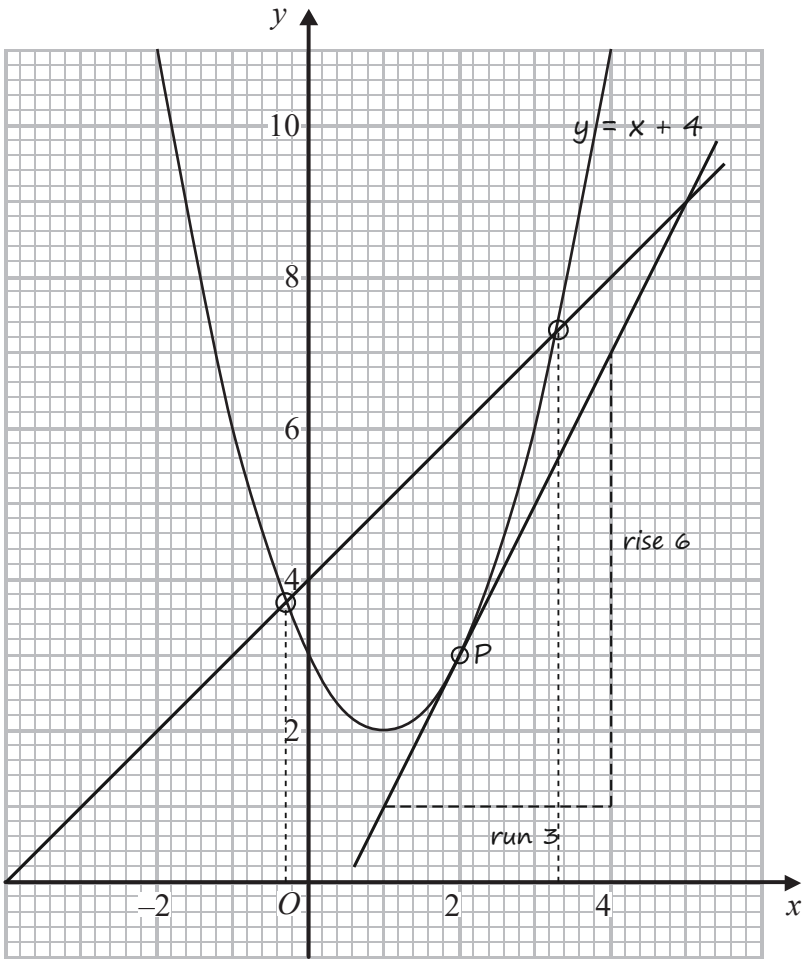
**20** The diagram shows part of the graph of $y = x^2 - 2x + 3$




(a) By drawing a suitable straight line, use your graph to find estimates for the solutions of $x^2 - 3x - 1 = 0$

$x^2 - 3x - 1 = 0$ so $x^2 - 2x + 3 = x + 4$

**1 mark**
Draw y = x + 4 on the original graph

-0.3 and 3.3

**1 mark**
Read both intersection x-coordinates

**(2)**

$P$ is the point on the graph of $y = x^2 - 2x + 3$ where $x = 2$

(b) Calculate an estimate for the gradient of the graph at the point $P$.

Tangent at P(2, 3): use (1, 1) and (4, 7).

**1 mark**
Tangent drawn at x = 2 on the original curve

$$\text{Gradient} = \frac{7 - 1}{4 - 1} = \frac{6}{3} = 2$$

**1 mark**
Use rise ÷ run from the tangent

2

**1 mark**
Gradient 2; accepted range 1.6–2.5

**(3)**

**(Total for Question 20 is 5 marks)**